Model 1: Rental Car RM with a Single Station (Flow Formulation)

Sets:

C = Set of car classes (Luxury, Mid-Sized, and Compact)
S = Set of booking slots

Parameters:

Δ = Number of days in the planning horizon
p_k = Price per instance of car rented for slot k
d_k = Demand for slot k
D_j = Set of booking slots with cars departing (renting out) on day j
A_j = Set of booking slots with cars arriving (returning) on day j
I_{h0} = Initial inventory of car class h
v_{hk} = 1 if car class h can satisfy demand for slot k, 0 otherwise

Decision Variables:

x_{hk} = Number of cars in class h used to fill slot k
I_{hj} = Inventory of car class h at end of business day j

Model:

$$\max \sum_{h \in C} \sum_{k \in S} p_k x_{hk} \tag{1}$$

Subject to:

$$\sum_{h \in C} x_{hk} \leq d_k \quad \forall k \in S \tag{2}$$

$$x_{hk} \leq M v_{hk} \quad \forall h \in C; k \in S \tag{3}$$

$$I_{hj} = I_{hj-1} - \sum_{k \in D_j} x_{hk} + \sum_{k \in A_j} x_{hk} \quad \forall h \in C; j = 1..\Delta \tag{4}$$

$$I_{hj}, x_{hk} \geq 0 \text{ and integer } \quad \forall h \in C; i \in L; j = 1..\Delta; k \in S \tag{5}$$

Model 2: Basic Multi-Site Rental Car RM

Sets:

C = Set of car classes (Luxury, Mid-Sized, and Compact)
L = Set of rental car station locations
S = Set of booking slots

Parameters:

Δ = Number of days in the planning horizon
p_k = Price per instance of car rented for slot k
d_k = Demand for slot k
$c_{hii'}$ = Transfer cost per car of class h from location i to i'
D_{ij} = Set of booking slots with cars departing from location i on day j
A_{ij} = Set of booking slots with cars arriving at location i on day j
I_{hi0} = Initial inventory of car class h at location i
v_{hk} = 1 if car class h can satisfy demand for slot k, 0 otherwise

Decision Variables:

x_{hk} = Number of cars in class h used to fill slot k
I_{hij} = Inventory of car class h at location i at end of business day j
$t_{hii'j}$ = Number of cars in class h transferred from location i to i' on night j

Model:

$$\max \sum_{h\in C}\sum_{k\in S} p_k x_{hk} - \sum_{h\in C}\sum_{i\in L}\sum_{i'\in L}\sum_{j=0}^{\Delta-1} c_{hii'}\, t_{hii'j} \tag{1}$$

Subject to:

$$\sum_{h\in C} x_{hk} \le d_k \;\; \forall k \in S \tag{2}$$

$$x_{hk} \le M v_{hk} \;\; \forall h \in C; k \in S \tag{3}$$

$$I_{hij} = I_{hij-1} - \sum_{i'\in L} t_{hii'j-1} + \sum_{i'\in L} t_{hi'ij-1} - \sum_{k\in D_{ij}} x_{hk} + \sum_{k\in A_{ij}} x_{hk} \;\; \forall h \in C; i \in L; j = 1..\Delta \tag{4}$$

$$\sum_{i'\in L} t_{hii'j} \le I_{hij} \;\; \forall h \in C; i \in L; j = 0..\Delta - 1 \tag{5}$$

$$I_{hij} \ge 0 \text{ and integer } \;\; \forall h \in C; i \in L; j = 1..\Delta \tag{6}$$

$$x_{hk}, t_{hii'j} \ge 0 \text{ and integer } \;\; \forall h \in C; i, i' \in L; j = 0..\Delta - 1 \tag{7}$$

Model 3: Multi-Site Rental Car RM with Realistic Transportation

Sets:

C = Set of car classes (Luxury, Mid-Sized, and Compact)
L = Set of rental car station locations
S = Set of booking slots

Parameters:

Δ = Number of days in the planning horizon
p_k = Price per instance of car rented for slot k
d_k = Demand for slot k
$c_{ii'}$ = Transfer cost per truck from location i to i'
D_{ij} = Set of booking slots with cars departing from location i on day j
A_{ij} = Set of booking slots with cars arriving at location i on day j
I_{hi0} = Initial inventory of car class h at location i
v_{hk} = 1 if car class h can satisfy demand for slot k, 0 otherwise
T_{i0} = Initial inventory of transport trucks at location i at end of day j
K^T = Capacity (cars) per transport truck

Decision Variables:

x_{hk} = Number of cars in class h used to fill demand slot k
$y_{ii'j}$ = Number of transport trucks sent from location i to i' on night j
I_{hij} = Inventory of car class h at location i at end of day j
$t_{hii'j}$ = Number of cars in class h transferred from location i to i' on night j
T_{ij} = Inventory of transport trucks at location i at end of day j

Model:

$$\max \sum_{h\in C}\sum_{k\in S} p_k x_{hk} - \sum_{i\in L}\sum_{i'\in L}\sum_{j=0}^{\Delta-1} c_{ii'} y_{ii'j} \tag{1}$$

Subject to:

$$\sum_{h\in C} x_{hk} \le d_k \ \ \forall k \in S \tag{2}$$

$$x_{hk} \le M v_{hk} \ \ \forall h \in C; k \in S \tag{3}$$

$$I_{hij} = I_{hij-1} - \sum_{i'\in L} t_{hii'j-1} + \sum_{i'\in L} t_{hi'ij-1} - \sum_{k\in D_{ij}} x_{hk} + \sum_{k\in A_{ij}} x_{hk} \ \ \forall h \in C; i \in L; j = 1..\Delta \tag{4}$$

$$\sum_{i'\in L} y_{ii'j} \le T_{ij} \ \ \forall i \in L; j = 0..\Delta - 1 \tag{5}$$

$$T_{ij} = T_{ij-1} - \sum_{i'\in L} y_{ii'j-1} + \sum_{i'\in L} y_{i'ij-1} \ \ \forall i \in L; j = 1..\Delta \tag{6}$$

$$\sum_{h\in C} t_{hii'j} \le K^T y_{ii'j} \ \ \forall i, i' \in L; j = 0..\Delta - 1 \tag{7}$$

$$I_{hij}, T_{ij} \ge 0 \text{ and integer } \ \forall h \in C; i \in L; j = 1..\Delta \tag{8}$$

$$x_{hk}, t_{hii'j}, y_{ii'j} \ge 0 \text{ and integer } \ \forall h \in C; i, i' \in L; j = 0..\Delta - 1 \tag{9}$$

Model 4: Multi-Site Rental Car RM with Realistic Transportation – Incorporating Dynamic Pricing and Fluctuating Demand

Sets:
C = Set of car classes (Luxury, Mid-Sized, and Compact)
L = Set of rental car station locations
S = Set of booking slots

Parameters:
Δ = Number of days in the planning horizon
p_k = Price per instance of car rented for demand slot k
d_k = Demand for slot k
d_k^c = Committed demand for slot k
$c_{ii'}$ = Transfer cost per truck from location i to i'
D_{ij} = Set of booking slots with cars departing from location i on day j
A_{ij} = Set of booking slots with cars arriving at location i on day j
I_{hi0} = Initial inventory of car class h at location i
$u_{kk'}$ = 1 if booking slots k and k' conflict, 0 otherwise
v_{hk} = 1 if car class h can satisfy demand for slot k, 0 otherwise
T_{i0} = Initial inventory of transport trucks at location i at end of day j
K^T = Capacity (cars) per transport truck

Decision Variables:
x_{hk} = Number of cars in class h used to fill demand slot k
$y_{ii'j}$ = Number of transport trucks sent from location i to i' on night j
I_{hij} = Inventory of car class h at location i at end of day j
$t_{hii'j}$ = Number of cars in class h transferred from location i to i' on night j
T_{ij} = Inventory of transport trucks at location i at end of day j

Model:

$$\max \sum_{h \in C} \sum_{k \in S} p_k x_{hk} - \sum_{i \in L} \sum_{i' \in L} \sum_{j=0}^{\Delta-1} c_{ii'} y_{ii'j} \tag{1}$$

Subject to:

$$d_k^c \leq \sum_{h \in C} x_{hk} \leq d_k \ \ \forall k \in S \tag{2}$$

$$x_{hk} \leq M v_{hk} \ \ \forall h \in C; k \in S \tag{3}$$

$$I_{hij} = I_{hij-1} - \sum_{i' \in L} t_{hii'j-1} + \sum_{i' \in L} t_{hi'ij-1} - \sum_{k \in D_{ij}} x_{hk} + \sum_{k \in A_{ij}} x_{hk} \ \ \forall h \in C; i \in L; j = 1..\Delta \tag{4}$$

$$\sum_{i' \in L} y_{ii'j} \leq T_{ij} \ \ \forall i \in L; j = 0..\Delta - 1 \tag{5}$$

$$T_{ij} = T_{ij-1} - \sum_{i' \in L} y_{ii'j-1} + \sum_{i' \in L} y_{i'ij-1} \ \ \forall i \in L; j = 1..\Delta \tag{6}$$

$$\sum_{h \in C} t_{hii'j} \leq K^T y_{ii'j} \ \ \forall i, i' \in L; j = 0..\Delta - 1 \tag{7}$$

$$I_{hij}, T_{ij} \geq 0 \text{ and integer } \ \forall h \in C; i \in L; j = 1..\Delta \tag{8}$$

$$x_{hk}, t_{hii'j}, y_{ii'j} \geq 0 \text{ and integer } \ \forall h \in C; i, i' \in L; j = 0..\Delta - 1 \tag{9}$$